

BOSTON LOGAN, MA, USA

WMO: 725090

Lat: 42.361N

Lon: 71.010W

Elev: 4

StdP: 101.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-10.6	-23.8	0.4	-11.9	-21.3	0.6	-8.9	14.2	2.6	12.7	-0.5	7.2	300	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	32.7	22.8	30.9	22.0	29.1	21.1	24.4	30.0	23.6	28.4	22.7	27.1	6.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.4	27.1	21.9	16.6	26.1	21.2	15.9	25.5	73.5	30.2	70.0	28.4	66.9	27.1	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.7	9.2	DB	-16.5	35.4	2.8	1.6	-18.5	36.6	-20.2	37.5	-21.7	38.5	-23.7	39.7	(4)
(5)			WB	-17.6	25.8	2.5	0.8	-19.4	26.4	-20.9	26.9	-22.3	27.3	-24.1	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.2	0.0	3.6	9.3	14.6	19.9	23.5	22.7	18.9	12.9	7.2	2.3		
	DBStd	9.53	5.64	5.02	4.81	4.32	4.22	4.09	3.23	3.04	3.53	3.96	4.47	4.78		
	HDD10.0	1269	347	282	207	62	5	0	0	0	0	17	107	243		
	HDD18.3	3054	604	514	458	272	132	29	2	2	34	174	334	497		
	CDD10.0	1707	1	1	8	42	148	298	418	393	266	106	22	4		
	CDD18.3	451	0	0	1	2	17	77	161	137	51	6	0	0		
	CDH23.3	3258	0	0	4	33	156	558	1272	929	275	30	1	0		
	CDH26.7	1067	0	0	1	11	49	183	457	288	73	4	0	0		
(14) Wind	WSAvg	5.0	5.6	5.5	5.6	5.4	4.8	4.5	4.4	4.3	4.5	4.9	5.0	5.3		
(15) Precipitation	PrecAvg	1077	88	87	100	90	83	89	77	81	87	91	100	104		
	PrecMax	1362	268	202	378	243	317	335	194	203	250	244	235	247		
	PrecMin	603	16	18	14	21	13	0	14	17	18	10	16	21		
	PrecStd	170	48	42	59	47	51	69	40	47	52	53	52	51		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	16.0	21.9	28.0	31.2	33.6	35.1	34.2	32.2	26.8	20.8	17.4		
		MCWB	13.2	12.3	13.4	16.9	19.9	23.1	24.4	23.9	22.8	18.7	16.3	14.2		
	2%	DB	12.1	11.9	16.2	22.1	27.6	31.1	32.8	31.7	28.6	23.3	18.0	13.8		
		MCWB	10.0	8.5	11.0	14.1	18.2	21.8	23.0	22.8	21.1	18.0	14.5	11.2		
	5%	DB	8.9	9.1	12.9	18.7	24.3	28.6	31.1	29.7	26.4	21.0	15.9	11.4		
		MCWB	6.4	5.9	8.3	12.3	16.7	20.5	22.3	21.8	20.0	16.7	12.3	8.7		
	10%	DB	6.2	6.7	10.1	15.9	21.4	26.6	29.3	28.0	24.3	18.8	13.8	9.1		
		MCWB	3.9	3.7	6.2	10.6	15.3	19.6	21.5	21.1	19.2	14.7	10.6	6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	13.3	15.2	18.1	21.5	24.6	25.7	25.4	23.9	21.2	17.7	14.8		
		MCDB	15.0	15.2	19.8	24.5	28.9	31.1	32.4	32.0	29.0	23.9	20.0	16.4		
	2%	WB	10.4	9.2	12.1	15.7	19.8	23.2	24.6	24.2	22.5	19.2	15.5	11.8		
		MCDB	11.6	11.3	15.4	20.7	25.5	29.0	30.4	29.2	26.3	22.1	17.5	13.5		
	5%	WB	6.8	6.3	9.0	13.6	18.0	22.1	23.7	23.4	21.5	17.4	13.1	9.3		
		MCDB	8.5	8.3	11.9	17.2	22.5	26.6	28.7	27.6	24.8	20.2	15.1	11.0		
	10%	WB	4.2	4.2	6.6	11.2	16.4	20.8	22.8	22.5	20.5	15.6	11.3	6.9		
		MCDB	5.8	6.2	9.3	15.1	20.7	24.7	27.3	26.3	23.6	18.1	13.2	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.4	7.4	8.1	8.2	8.2	8.1	7.8	7.6	7.5	7.0	6.7		
		MCDBR	10.2	10.3	11.9	13.2	13.3	11.4	10.7	10.3	10.4	10.3	9.5	9.3		
	5% WB	MCWBR	9.4	8.2	8.2	7.5	6.5	5.5	4.4	4.4	5.2	6.5	7.5	8.5		
		MCWBR	10.4	9.6	11.5	12.1	11.6	10.7	9.5	9.2	8.9	9.5	8.6	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.302	0.325	0.370	0.406	0.452	0.472	0.434	0.376	0.335	0.311	0.295		
	taud		2.505	2.493	2.487	2.392	2.329	2.234	2.185	2.278	2.423	2.543	2.571	2.568		
	Ebn at Noon		861	909	923	899	870	827	805	826	856	859	834	830		
	Edh at Noon		71	85	96	114	125	138	143	127	102	80	67	63		
(44) All-Sky Solar Radiation	RadAvg		1.76	2.63	3.63	4.67	5.20	5.62	5.92	5.32	4.28	2.84	1.93	1.46		
	RadStd		0.11	0.22	0.25	0.50	0.58	0.53	0.36	0.24	0.29	0.24	0.15	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+48		
(47) Regional (72 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	+42		

Nomenclature: See separate page